

1. If a given system loses 2736 J of internal energy with no exchange of heat while expanding from a volume of 5.0 to 15.0 L, then against what external pressure (in atm) did the expansion occur?

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- A) 1.2 atm
- B) -2.7 atm
- C) 2.7 atm
- D) 270 atm
- E) -270 atm

2. Which statement is true regarding enthalpy?

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- A) ΔH is equal to q at constant volume.
- B) ΔH is a value dependent on the path taken by the system.
- C) At constant volume, ΔH is zero.
- D) ΔH can be determined using constant pressure calorimetry or using enthalpies of formation.
- E) ΔH is equal to change in internal energy at constant pressure.

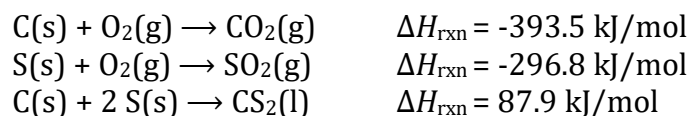
3. The temperature of a given bomb calorimeter (constant volume calorimeter) increases by 0.10 K when 7.52 J of heat is added. The temperature of the same calorimeter increases by 25 K when 30.0 g of a certain fuel is combusted in the calorimeter. What is the enthalpy of combustion for the fuel in J?

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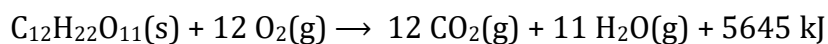
- A) 0.63 J
- B) -23 J
- C) -560 J
- D) -1900 J
- E) 56,000 J

4. Find ΔH_{rxn} for the reaction $\text{CS}_2(\text{l}) + 3 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2 \text{SO}_2(\text{g})$ given that:



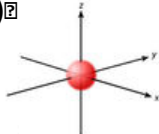
- A) -1196.0 kJ/mol
- B) -1075.0 kJ/mol
- C) -899.2 kJ/mol
- D) -602.4 kJ/mol
- E) -454.0 kJ/mol

5. How many moles of oxygen gas are required to produce 9500 kJ of energy in the presence of excess sucrose ($C_{12}H_{22}O_{11}$)? You may assume 100% yield for the reaction.

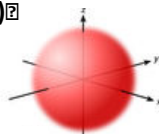


- A) 1.7 mol
B) 7.1 mol
C) 20. mol
D) 45 mol
E) 790 mol
6. Which statement is true regarding electromagnetic radiation?
- A) Photon A has greater frequency than photon B; therefore, photon A has a longer wavelength than photon B.
B) Gamma rays are less energetic than visible light.
C) The frequency of blue light is greater than that of red light.
D) Radio waves have shorter wavelengths than x-rays.
E) Increasing the amplitude of a wave increases its energy.
7. The three quantum numbers for an electron in a hydrogen atom in a certain state are $n = 4$, $l = 2$, and $m_l = -2$. Which of these orbitals best represents the orbital in which the electron is located (Note: all are drawn with approximately the same distance scales)?

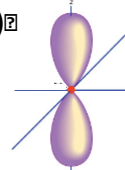
A)



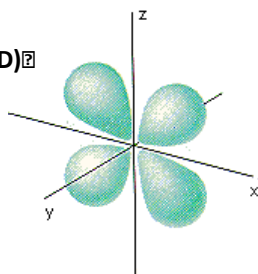
B)



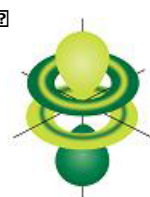
C)



D)



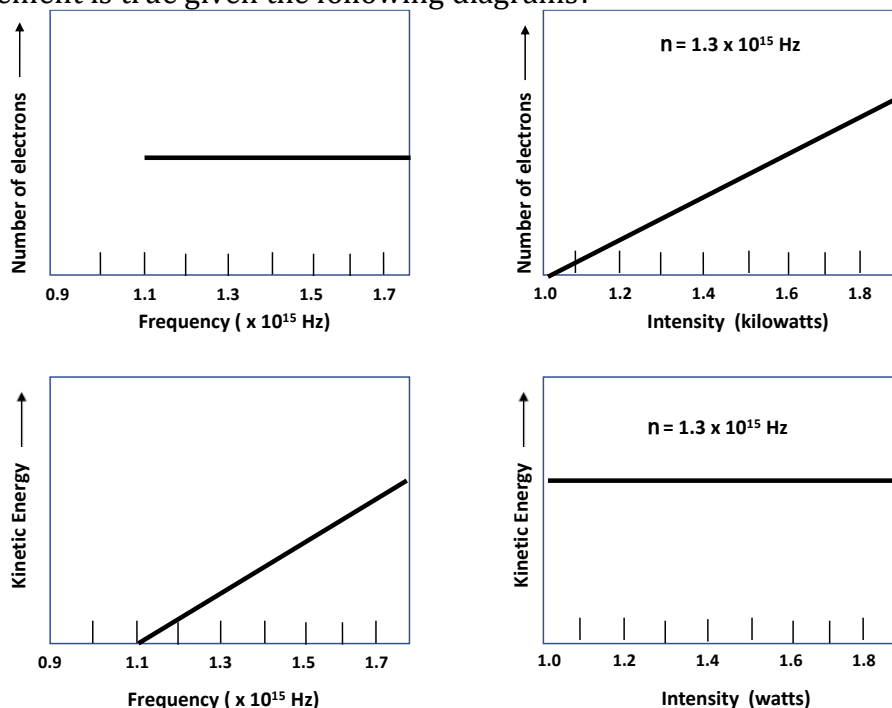
E)



8. Which is the highest energy orbital that is occupied in the ground-state electron configuration uranium (U) ?

- A) $6d$
- B) $7f$
- C) $4d$
- D) $5f$
- E) $9f$

9. Which statement is true given the following diagrams?



- A) The threshold frequency for the metal is 1.3×10^{15} Hz.
- B) More electrons are ejected when the frequency of the incident radiation is 1.5×10^{15} Hz than when it is 0.9×10^{15} Hz.
- C) Increasing the intensity of the incident radiation increases the kinetic energy of the ejected electrons.
- D) The graphs can be interpreted adequately by treating light as a wave.
- E) Increasing the kinetic energy of the incident radiation increases the number of electrons ejected.

10. Which statement(s) is/are true regarding atomic emission spectra?

- I. Atomic emission spectra are continuous, indicating that electrons can have any energy.
- II. The wavelength of light emitted when a photon transitions from $n = 4$ to $n = 3$ is longer than the wavelength emitted when a photon transitions from $n = 2$ to $n = 1$.
- III. Atomic emission spectra are discrete which provides evidence for the quantization of electrons.

- A) I only
- B) II only
- C) III only
- D) II and III
- E) I and III

11. Which statement is an application or result of Hund's rule?

- A) The 5s orbital must be filled before the 4d orbitals.
- B) There are only two allowable values for the spin quantum number.
- C) Manganese has five unpaired electrons in its ground-state electron configuration.
- D) Transition metals lose their ns electrons before their $(n-1)d$ electrons.
- E) For hydrogen, the 3s, 3p, and 3d orbitals all are degenerate.

12. Which statement is true when sulfur forms the sulfide ion?

- A) Two electrons are lost from 2p orbitals.
- B) Two electrons are added to the 3p orbitals.
- C) Sulfur becomes argon.
- D) Two electrons are added to the 3d orbitals.
- E) Two electrons are lost from the 2d orbitals.

13. Which statement is true regarding atomic radii?

- A) The atomic radius for Si is greater than for C because Si has more core electrons and its valence electrons occupy higher energy orbitals.
- B) The atomic radius for B is greater than for Li because B's effective nuclear charge is closer to its actual nuclear charge than is Li's.
- C) The atomic radius for F is smaller than for N because N is an exception to the trend as a result of its half-full set of *p* orbitals.
- D) The atomic radius for Ca is smaller than for Sr because Sr has more valence electrons.
- E) The atomic radius for Na is smaller than for Mg because Mg has greater nuclear charge.

14. Which answer option presents feasible identities for A, B, C, and/or D given the image below?



- A) Sphere A represents Cl, and sphere C represents Cl⁺.
- B) Sphere A represents Na, and sphere C represents Na⁻.
- C) Sphere B represents Na, and sphere D represents Na⁺.
- D) Sphere B represents Na, and sphere D represents Na⁻.
- E) Sphere A represents Cl, and sphere C represents Na⁺.

15. An unknown element gives rise to a common ion with the same electron configuration as Ne. The ionic radius of this element is less than the atomic radius. Given the ionization energies below for the element, which statement is true regarding the element ?

<u>Ionization Number</u>	<u>Energy (kJ/mol)</u>
IE ₁	737.7
IE ₂	1450.6
IE ₃	7732.8
IE ₄	10987.6
IE ₅	14533.9
IE ₆	19450.5
IE ₇	24367.2

- A) The element is a nonmetal.
B) The element has a very negative electron affinity.
C) The element is sulfur.
D) The element is more likely to gain electrons than to lose them.
E) The element's electron affinity is greater than zero.
16. Which atom or ion has the largest radius?
- A) P³⁻
B) S²⁻
C) Cl⁻
D) K⁺
E) Ca²⁺
17. Which bond can best be classified as polar covalent?

- A) Ca—Cl
B) P—Cl
C) C—C
D) C—H
E) Li—O

18. The Lewis structure for which molecule contains at least one double bond? ?

- I. CO_2
- II. XeF_4
- III. SiS_2

- A) I only
- B) II only
- C) III only
- D) Both I and III
- E) All of I, II, and III

19. Which statement is true regarding the Lewis structure for SF_4 ?

- A) The structure has a central atom with four single bonds and no lone pairs of electrons.
- B) The structure has one double bond and three single bonds.
- C) Sulfur has exactly an octet of electrons.
- D) The structure has four bonding and one lone pair of electrons on the central atom.
- E) More than one of these statements is true.

20. Which ionic solid would you expect to have the greatest magnitude of lattice energy?

- A) NaCl
- B) Na_2S
- C) CaCl_2
- D) CaO
- E) LiBr

21. Which form of the exam do you have?

- A)
- B)